

Learn
Practice
Build
Grow →

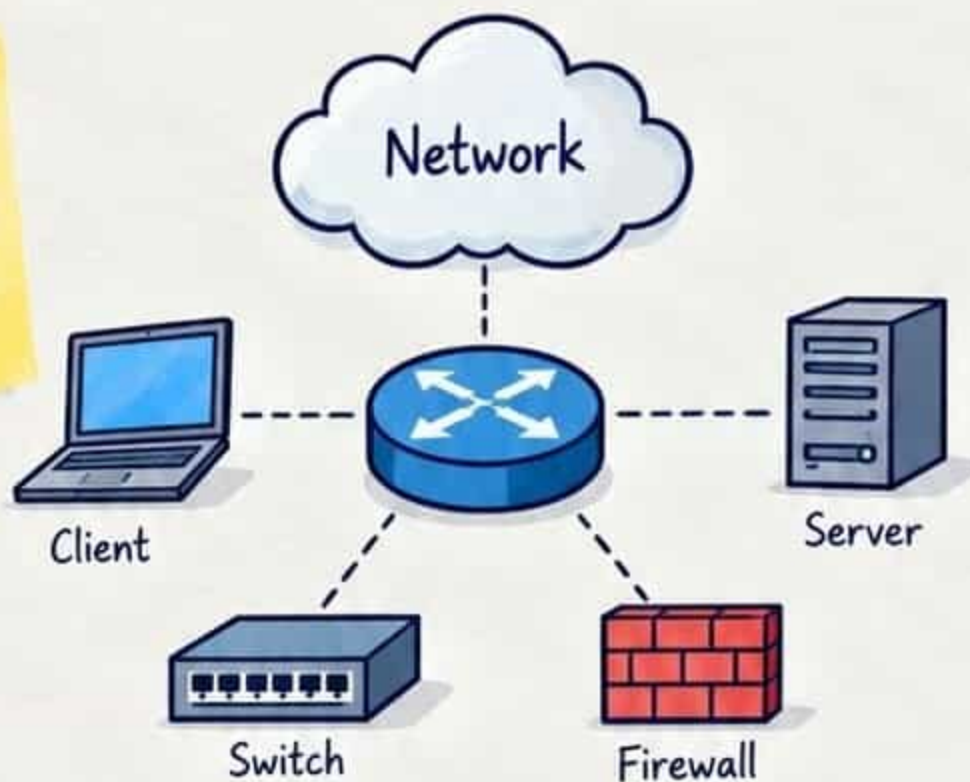
"Networks
Connect
a Better
World"

CCNA Notes

Complete Notes for CCNA (200-301)

Simple | Structured | Exam Focused

From
Basics
to
Confidence



Good
Networks
Build
Great
Opportunities



Concepts
Explained



Diagrams
& Examples



Key Commands
& Configurations



Exam Tips
& Quick Revision

1. Networking Fundamentals

What is a Computer Network?

- A computer network is a group of interconnected devices that can communicate and share resources (data, applications, hardware) with each other.
- It can be small (home network) or large (global network - Internet).
- Networks help in sharing information, improving communication and increasing efficiency.

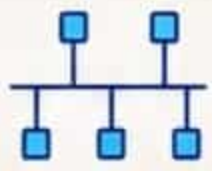
Key Benefits

- ✓ Share data and resources
- ✓ Enable communication
- ✓ Improve productivity
- ✓ Centralized management
- ✓ Access to cloud and internet services

Types of Networks (by size/coverage)

Type	Full Form	Coverage Area	Example
PAN	Personal Area Network	Few meters	Bluetooth, Tethering
LAN	Local Area Network	Home, Office, Building	Office network
MAN	Metropolitan Area Network	City	City-wide network
WAN	Wide Area Network	Country / Global	Internet

Network Topologies

Star	Bus	Ring	Mesh
 - Most common - Easy to manage - If one device fails, others work	 - Simple & low cost - All devices share the same cable - Not commonly used now	 - Each device connected to two others - Data flows in one direction	 - High redundancy - More reliable - Expensive - Used in large networks

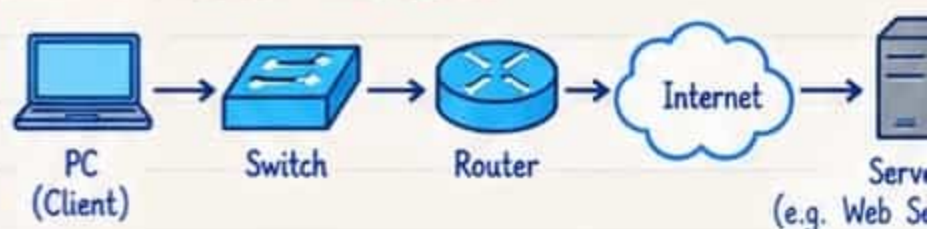
Network Devices and Their Functions

Device	Function	Symbol
Router	Connects different networks and routes data between them.	
Switch	Connects devices within a LAN and forwards data using MAC addresses.	
Access Point	Provides wireless connectivity to devices.	
Firewall	Monitors and controls network traffic, blocks unwanted traffic.	
Server	Provides resources and services (files, applications, web, email, etc.).	
End Device	Client devices like PC, laptop, phone, printer, etc. that use network resources.	

LAN vs WAN

Feature	LAN	WAN
Coverage	Small area (building/office)	Large area (country/global)
Speed	High (100 Mbps - 10 Gbps)	Lower (compared to LAN)
Ownership	Private	Service provider (ISP)
Cost	Lower	Higher
Example	Office network	Internet

Basic Network Diagram



How a Network Works?

- A device (client) sends a request for data or a service.
- Data is sent through switches/routers towards the destination.
- Routers select the best path to forward the data.
- The destination device (server) receives the request and sends a response.
- Data is delivered back to the client.

Important Terms

Term	Meaning
IP Address	Logical address of a device (e.g. 192.168.1.10)
MAC Address	Physical address of a device (e.g. 00:1A:2B:3C:4D:5E)
DNS	Converts domain name to IP address
DHCP	Automatically assigns IP addresses
Gateway	Device that connects local network to other networks
Bandwidth	Maximum data transfer capacity
Latency	Time taken for data to travel from source to destination

2. OSI Model & TCP/IP Model

OSI and TCP/IP models are reference models that help us understand how data is transmitted across a network from one device to another.

OSI Model (7 Layers)

Layer	Name	Functions	Examples
7	Application	Provides network services to end users	HTTP, HTTPS, FTP, SMTP, DNS
6	Presentation	Data formatting, encryption, compression	SSL/TLS, JPEG, MPEG
5	Session	Manages sessions between devices	NetBIOS, RPC
4	Transport	End-to-end communication, reliability, flow control	TCP, UDP
3	Network	Logical addressing, routing and path selection	IP, ICMP, OSPF
2	Data Link	Framing, error detection, MAC addressing	Ethernet, PPP, Switching
1	Physical	Transmits raw bits over physical media	Cables, Hubs, Radio, Fiber

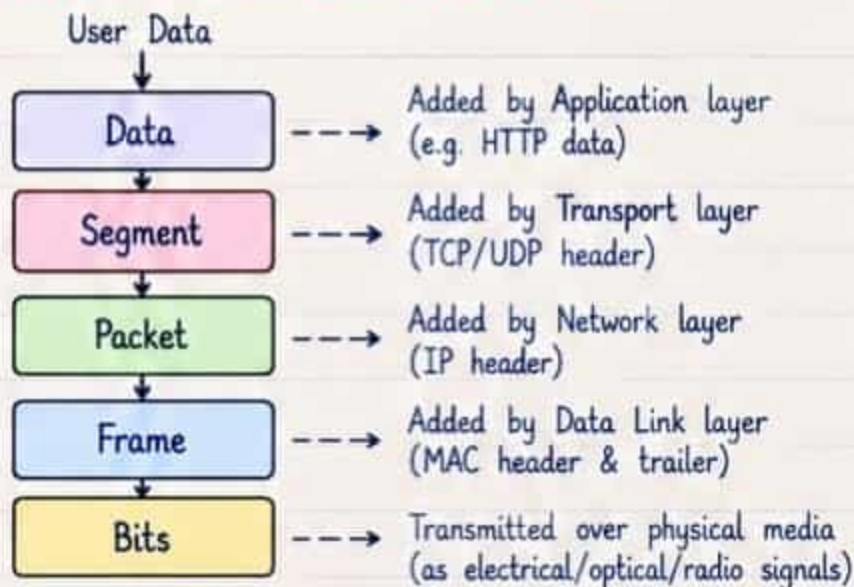
TCP/IP Model (4 Layers)

Layer	Name	Functions	Equivalent OSI Layers
4	Application	Provides network services to user applications	7, 6, 5
3	Transport	End-to-end communication (reliability, flow control)	4
2	Internet	Logical addressing and routing	3
1	Network Access	Framing, MAC addressing and physical transmission	2, 1

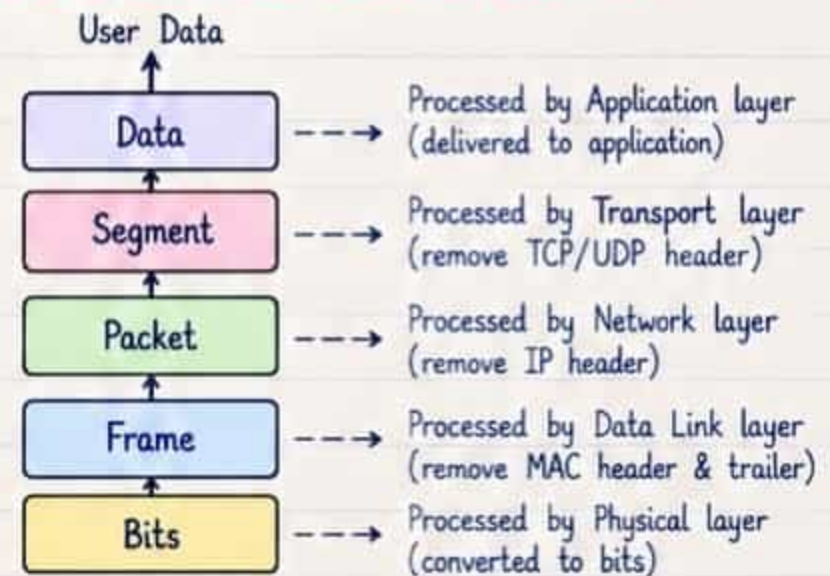
Key Differences

- OSI is a theoretical model (7 layers) by ISO.
- TCP/IP is a practical model (4 layers) used in real networks.
- TCP/IP combines Application, Presentation and Session layers.
- OSI model is used for learning and troubleshooting.
- TCP/IP model is used in real implementations (Internet).

Encapsulation (Sending Data)



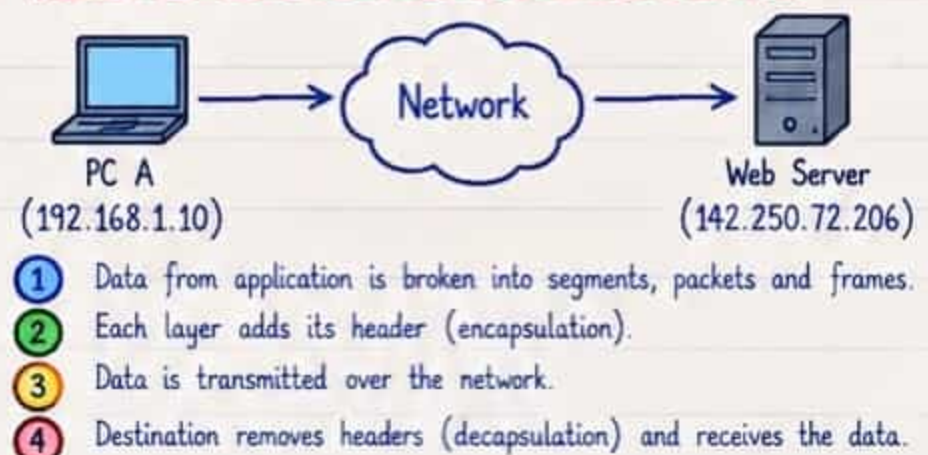
Decapsulation (Receiving Data)



PDU (Protocol Data Unit)

Layer	PDU Name	Example
Application	Data	HTTP Data, DNS Query
Transport	Segment (TCP) / Datagram (UDP)	TCP Segment, UDP Datagram
Network	Packet	IP Packet
Data Link	Frame	Ethernet Frame
Physical	Bits	0s and 1s

End-to-End Communication Example



3. Ethernet, MAC Address & ARP

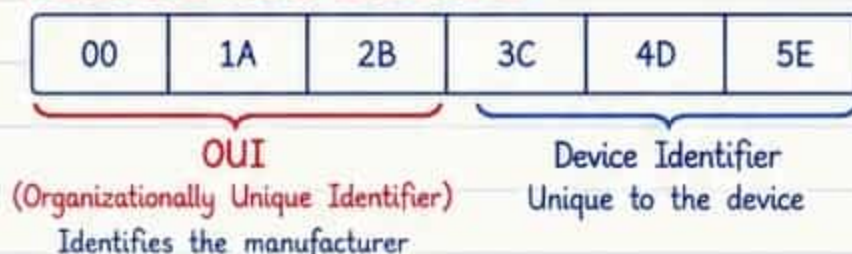
What is Ethernet?

- Ethernet is the most common LAN technology used to connect devices in a network.
- It defines how data is framed and sent over physical media (copper or fiber).
- Common Ethernet speeds: 10 Mbps, 100 Mbps, 1 Gbps, 10 Gbps and higher.
- Uses MAC addresses to deliver data within the local network.

MAC address

- A MAC (Media Access Control) address is a unique hardware address assigned to a network interface card (NIC).
- It is 48 bits long (6 bytes) and usually written in hexadecimal format.
- Example: 00:1A:2B:3C:4D:5E

MAC address structure



Types of MAC addresses

Type	Description	First Octet (Binary)	Example
Unicast	Identifies a single device (one-to-one).	0xxxxxxx (LSB = 0)	00:1A:2B:3C:4D:5E
Multicast	Identifies a group of devices (one-to-many).	1xxxxxxx (LSB = 1)	01:00:5E:00:00:01
Broadcast	Identifies all devices in the local network.	All 1s (FF)	FF:FF:FF:FF:FF:FF

How a switch learns MAC addresses

1	Receive a frame on a port.
2	Read the source MAC address.
3	Add the source MAC and port to the MAC address table.
4	Forward the frame based on the destination MAC address.
5	If destination is unknown, flood the frame out all other ports.
6	Over time, the switch builds a MAC address table.

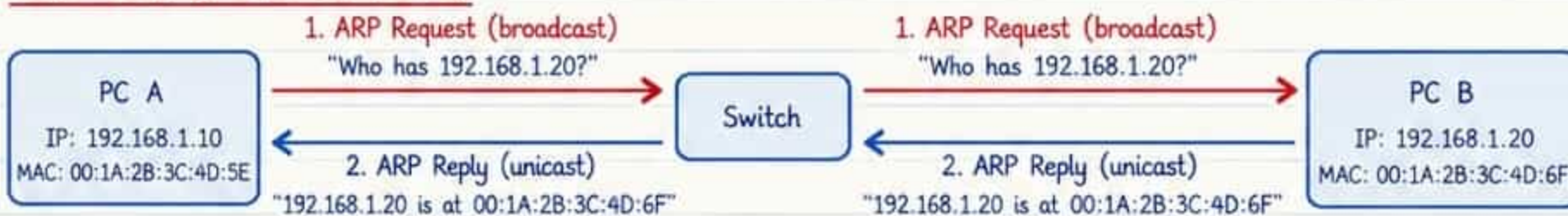
MAC address table (example)

MAC Address	Port	Type
00:1A:2B:3C:4D:5E	Fa0/1	Dynamic
00:1A:2B:3C:4D:6F	Fa0/2	Dynamic
00:1A:2B:3C:4D:7A	Fa0/3	Dynamic

ARP (Address Resolution Protocol)

- ARP maps an IP address to a MAC address within a local network.
- When a device knows the IP address but not the MAC address, it sends an ARP request (broadcast).
- The device with the matching IP address responds with an ARP reply (unicast).

ARP flow (PC A to PC B)



Hub vs Switch

Feature	Hub	Switch
Operation	Layer 1 (Physical)	Layer 2 (Data Link)
How it forwards	Sends data out all ports (no learning)	Forwards based on MAC address table
Collision domains	One shared collision domain	One collision domain per port
Performance	Lower (more collisions)	Higher (reduces collisions)
Use case	Legacy / rarely used	Standard in modern networks

Remember:

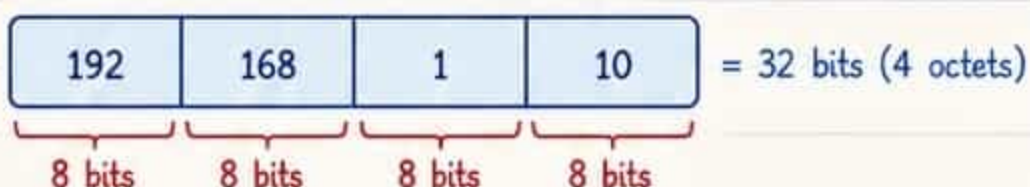
Ethernet uses MAC addresses for local delivery, switches learn MAC addresses, and ARP resolves IP addresses to MAC addresses.

4. IPv4 Addressing

What is IPv4?

- IPv4 is a 32-bit addressing system used to identify devices on a network.
- It is written as four octets (8 bits each) in decimal format, separated by dots.
- Example: 192.168.1.10 (each octet can be 0 – 255)

IPv4 Address Structure (32 bits)



Subnet Mask & CIDR

- Subnet mask tells which part is the network and which part is the host.
- CIDR (Classless Inter-Domain Routing) uses a "/" notation to show the number of network bits.
- Example: 192.168.1.10/24 means the first 24 bits are the network portion.

Common IPv4 Address Types

Type	Range	Purpose / Notes
Private	10.0.0.0 – 10.255.255.255 172.16.0.0 – 172.31.255.255 192.168.0.0 – 192.168.255.255	Used in internal networks (not routable on Internet).
Public	All other addresses	Used on the Internet (globally routable).
Loopback	127.0.0.0 – 127.255.255.255	Used for local testing (e.g. 127.0.0.1).
APIPA	169.254.0.0 – 169.254.255.255	Automatically assigned when no DHCP server is available.

Example: 192.168.1.10/24

Item	Value	Explanation
IP Address	192.168.1.10/24	Host address with 24 network bits.
Subnet Mask	255.255.255.0	/24 = 24 ones in the mask.
Network Address	192.168.1.0	First address in the subnet.
Usable Host Range	192.168.1.1 – 192.168.1.254	Addresses available for hosts.
Broadcast Address	192.168.1.255	Last address in the subnet (used for broadcast).
Total Addresses	256	$2^{(32-24)} = 256$ (254 usable hosts).

CIDR Quick View

CIDR	Subnet Mask	Total Addresses	Usable Hosts
/8	255.0.0.0	16,777,216	16,777,214
/16	255.255.0.0	65,536	65,534
/24	255.255.255.0	256	254
/25	255.255.255.128	128	126
/26	255.255.255.192	64	62
/27	255.255.255.224	32	30
/28	255.255.255.240	16	14
/29	255.255.255.248	8	6
/30	255.255.255.252	4	2

Key Points

- IPv4 uses 32 bits (4 octets).
- Subnet mask identifies network vs host portion.
- CIDR shows the number of network bits (e.g. /24).
- Private IP addresses are not routable on the Internet.
- Network address = first address in the subnet.
- Broadcast address = last address in the subnet.
- Usable hosts = total addresses minus 2 (network and broadcast).

Remember:

IP addressing is the foundation of communication in a network. Understand how IPv4 works – it's essential for every CCNA topic.

5. Subnetting Made Simple

Why is subnetting needed?

- Divides a large network into smaller networks (subnets).
- Uses IP addresses more efficiently.
- Improves network performance (smaller broadcast domains).
- Increases security by isolating networks (e.g. departments).
- Required for scalable and organized network design.

Subnet Cheat Table (Common Prefixes)

Prefix	Subnet Mask	Total Addresses	Usable Addresses	Block Size (Increment)
/24	255.255.255.0	256	254	256
/25	255.255.255.128	128	126	128
/26	255.255.255.192	64	62	64
/27	255.255.255.224	32	30	32
/28	255.255.255.240	16	14	16
/29	255.255.255.248	8	6	8
/30	255.255.255.252	4	2	4

Example: Split 192.168.1.0/24 into /26 subnets

Subnet #	Network Address	Usable Host Range	Broadcast Address
1	192.168.1.0/26	192.168.1.1 – 192.168.1.62	192.168.1.63
2	192.168.1.64/26	192.168.1.65 – 192.168.1.126	192.168.1.127
3	192.168.1.128/26	192.168.1.129 – 192.168.1.190	192.168.1.191
4	192.168.1.192/26	192.168.1.193 – 192.168.1.254	192.168.1.255

Fast Method for Subnetting

- 1 Find the block size.
Block size = 256 – last octet of subnet mask.
- 2 List the subnet network addresses.
Start at 0 and add the block size each time.
- 3 Find the usable host range.
Network address + 1 to Broadcast address – 1.
- 4 Find the broadcast address.
Network address + (block size – 1).
- 5 Repeat for all required subnets.

Example (block size calculation)

For /26 → subnet mask 255.255.255.192

Block size = 256 – 192 = 64

Subnets start at: 0, 64, 128, 192



Remember: Subnetting helps you design efficient, secure and scalable networks.
Know how to calculate the block size and find network, host and broadcast addresses!

6. VLSM & Subnetting Practice

What is VLSM?

- VLSM (Variable Length Subnet Masking) allows different subnet sizes (prefix lengths) within the same network.
- It breaks a larger network into smaller subnets of different sizes based on actual requirements.

Why is VLSM useful?

- More efficient use of IP address space.
- Reduces wastage compared to fixed subnetting.
- Supports networks with different host requirements (LANs, WANs, etc).
- Widely used in real networks to save addresses and simplify management.

Example Requirement

We have the network 192.168.10.0/24 and need to create subnets for:

- LAN A – 100 hosts
- LAN B – 50 hosts
- LAN C – 20 hosts
- WAN – 2 hosts

Step 1: Choose the right prefix for each requirement

Network	Required hosts	Usable hosts needed	Chosen prefix	Total addresses	Usable hosts
LAN A	100	100	/25	128	126
LAN B	50	50	/26	64	62
LAN C	20	20	/27	32	30
WAN	2	2	/30	4	2

Step 2: Allocate subnets (from largest to smallest)

Network	Network address	Prefix	Usable host range	Broadcast address
LAN A	192.168.10.0	/25	192.168.10.1 – 192.168.10.126	192.168.10.127
LAN B	192.168.10.128	/26	192.168.10.129 – 192.168.10.190	192.168.10.191
LAN C	192.168.10.192	/27	192.168.10.193 – 192.168.10.222	192.168.10.223
WAN	192.168.10.224	/30	192.168.10.225 – 192.168.10.226	192.168.10.227

Fixed Subnetting vs VLSM

Feature	Fixed Subnetting	VLSM
Subnet size	All subnets are the same size	Different subnet sizes based on need
Address usage	Can waste many addresses	More efficient, less wastage
Flexibility	Less flexible	More flexible
Use case	Simple networks, older designs	Modern networks (real-world)

Key Takeaway: VLSM helps you use IP addresses efficiently by giving each network the right size. Always allocate subnets from largest to smallest.

7. IPv6 Fundamentals

Why IPv6 is needed?

- IPv4 addresses are running out due to the growth of devices (IoT, mobile, cloud, etc).
- IPv6 provides a much larger address space (128-bit) to support future growth.
- It improves network efficiency, security (built-in IPsec) and simplifies address configuration.
- It removes the need for NAT in most cases (end-to-end connectivity).

Example IPv6 address

2001:0DB8:AC10:0000:0000:0000:0000:1A2B

IPv6 address simplification rules

- Leading zeros in each hextet can be removed.
- One or more consecutive hextets of all zeros can be replaced with :: (double colon) once per address.
- :: can only be used once in an address.
- Hexadecimal letters can be uppercase or lowercase (e.g. A-F or a-f).
- Example (simplified): 2001:db8:ac10::1a2b

Common IPv6 address types

Type	Prefix (Range)	Purpose
Global Unicast	2000::/3	Routable on the Internet, unique addresses (like public IPv4).
Link-Local	FE80::/10	Used for communication on the local link only (not routed).
Unique Local	FC00::/7	Private addresses for internal networks (like RFC1918 in IPv4).
Multicast	FF00::/8	One-to-many communication (replaces broadcast).
Loopback	::1/128	Refers to the local device (like 127.0.0.1 in IPv4).

IPv4 vs IPv6 comparison

Feature	IPv4	IPv6
Address size	32-bit	128-bit
Address format	Dotted decimal (e.g. 192.168.1.1)	Hexadecimal (e.g. 2001:db8::1)
Address space	~4.3 billion	~ 3.4×10^{38}
NAT	Commonly required	Not required (end-to-end)
Broadcast	Supported	Not used (multicast instead)
Security	Optional (IPsec)	Built-in support for IPsec
Configuration	Manual or DHCP	SLAAC, DHCPv6 or manual

Cisco configuration example

```
R1# configure terminal
R1(config)# interface g0/0
R1(config-if)# ipv6 address 2001:db8:ac10::1/64
R1(config-if)# no shutdown
```

Remember: IPv6 solves the address shortage problem and enables a more scalable, secure and efficient network for the future.

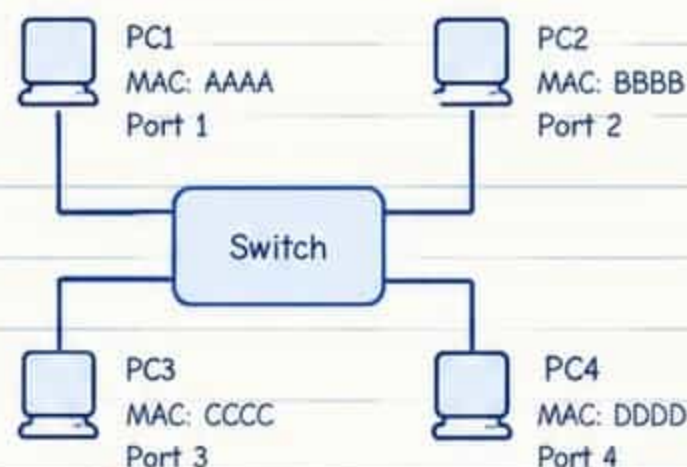
8. Switching Fundamentals

How a switch forwards frames?

- A switch operates at Layer 2 and forwards Ethernet frames based on MAC addresses.
- It examines the destination MAC address in the frame.
- It looks up the destination MAC in its MAC address table.
- If a match is found, it forwards the frame out the correct port (unicast).
- If no match is found, it floods the frame out all other ports (except the incoming port).

How a switch learns MAC addresses?

- A switch learns MAC addresses from the source MAC address of incoming frames.
- It adds the source MAC and the associated port to its MAC address table.
- MAC entries age out after a period of inactivity (default ~300 seconds on Cisco).
- This process is automatic and requires no manual configuration.



MAC Address Table (Example)

MAC Address	VLAN	Port	Type
0011.2233.4455	1	Fa0/1	Dynamic
00AA.BBCC.DDEE	1	Fa0/2	Dynamic
00FF.1122.3344	1	Fa0/3	Dynamic
AABB.CCDD.EEFF	1	Fa0/4	Dynamic

Frame Types and Switching Action

Frame Type	Switching Action
Known Unicast	Forward out the specific port associated with the destination MAC.
Unknown Unicast	Flood out all ports except the incoming port.
Broadcast	Flood out all ports except the incoming port.
Multicast	Forward out all ports in the multicast group (or flood if unknown).

Layer 2 vs Layer 3 Switch

Feature	Layer 2 Switch	Layer 3 Switch
Primary function	Switching (MAC address based)	Switching + Routing (MAC & IP based)
Operates at	Layer 2 (Data Link)	Layer 2 and Layer 3
Routing capability	No	Yes (inter-VLAN routing)
Typical use case	Access layer switching	Distribution/core, VLAN routing
Examples	Catalyst 2960 (L2)	Catalyst 3560/3850 (L3)

Useful Show Commands

Command	Purpose
show mac address-table	View the MAC address table.
show interfaces status	See interface status (up/down, VLAN, speed).
show interfaces	Detailed interface information (counters, errors, etc.).
show running-config	View current switch configuration.

Remember: A switch learns MAC addresses, builds a MAC table, and forwards frames efficiently. Unknown destinations are flooded, not dropped.

9. VLANs & Trunking

What is a VLAN?

- A VLAN (Virtual Local Area Network) logically separates a physical network into multiple broadcast domains.
- Devices in the same VLAN can communicate with each other, even if they are on different switches.

Why are VLANs used?

- Improve security by isolating departments (e.g. HR, Finance, Guest).
- Reduce broadcast traffic and improve performance.
- Provide logical separation without needing multiple physical networks.
- Make network management simpler and more flexible.

Access Port vs Trunk Port

Feature	Access Port	Trunk Port
Purpose	Connects to end devices (PC, IP phone)	Carries multiple VLANs between switches
VLANs carried	Single VLAN	Multiple VLANs
Frame format	Untagged	Tagged (802.1Q)
Typical use	User devices	Switch-to-switch, switch-to-router
Example	access vlan 10	switchport mode trunk

802.1Q VLAN Tagging

Destination MAC (6 bytes)	Source MAC (6 bytes)	VLAN Tag (4 bytes)	Data (46 - 1500 bytes)	FCS (4 bytes)
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802.1Q tag (contains VLAN ID, priority, etc.)

Example: Create VLAN and Assign Access Port

```
Switch(config)# vlan 10
Switch(config-vlan)# name Users
Switch(config-vlan)# exit
Switch(config)# interface fa0/1
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
Switch(config-if)# no shutdown
```

Example: Trunk Port Allowing Multiple VLANs

```
Switch(config)# interface fa0/24
Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk allowed vlan 10,20,30
Switch(config-if)# no shutdown
```

Important Terms

Term	Description
Native VLAN	VLAN for untagged traffic on a trunk link (default is VLAN 1).
Voice VLAN	Used for IP phones to separate voice traffic from data traffic.
Default VLAN	VLAN 1 (exists by default, all ports are in this VLAN initially).

Remember: VLANs help you build a more secure, efficient and manageable network. Use trunks to carry multiple VLANs between network devices.

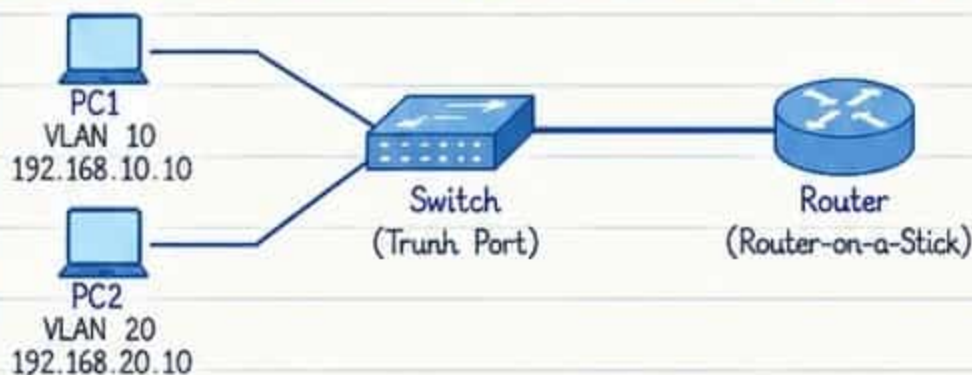
10. Inter-VLAN Routing

Why is Inter-VLAN Routing Needed?

- Devices in different VLANs are in different broadcast domains and cannot communicate by default.
- Inter-VLAN routing allows devices in different VLANs to communicate with each other.
- It is required when users or services in different departments (VLANs) need to access shared resources such as servers, printers or the internet.

Method 1: Router-on-a-Stick (RoAS)

- Uses a single physical interface on a router with multiple subinterfaces (one per VLAN).
- Each subinterface is configured with 802.1Q encapsulation for the respective VLAN ID.
- The router performs the routing between VLANs.
- The switch port connected to the router must be a trunk.



Router Configuration (RoAS)

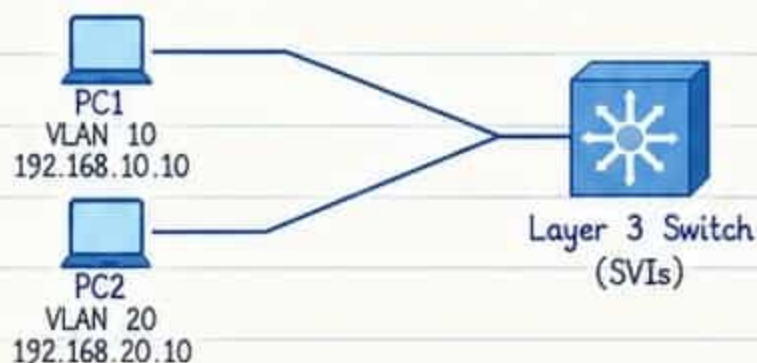
```
R1(config)# interface g0/0
R1(config-if)# no shutdown

R1(config)# interface g0/0.10
R1(config-subif)# encapsulation dot1Q 10
R1(config-subif)# ip address 192.168.10.1 255.255.255.0

R1(config)# interface g0/0.20
R1(config-subif)# encapsulation dot1Q 20
R1(config-subif)# ip address 192.168.20.1 255.255.255.0
```

Method 2: Layer 3 Switch (SVIs)

- Uses a Layer 3 switch with Switch Virtual Interfaces (SVIs).
- Each VLAN has an SVI with an IP address (default gateway for that VLAN).
- The switch performs inter-VLAN routing in hardware.
- **ip routing** must be enabled on the switch.



Layer 3 Switch Configuration (SVIs)

```
SW1(config)# ip routing

SW1(config)# interface vlan 10
SW1(config-if)# ip address 192.168.10.1 255.255.255.0
SW1(config-if)# no shutdown

SW1(config)# interface vlan 20
SW1(config-if)# ip address 192.168.20.1 255.255.255.0
SW1(config-if)# no shutdown
```

Comparison: Router-on-a-Stick vs Layer 3 Switch

Feature	Router-on-a-Stick (RoAS)	Layer 3 Switch (SVIs)
Device used	Router	Layer 3 Switch
Interfaces	Subinterfaces on a single physical interface	SVIs (one per VLAN)
Performance	Lower (software-based routing)	Higher (hardware-based routing)
Scalability	Good for small networks	Better for large networks
Configuration	Requires 802.1Q encapsulation	Simple SVI configuration + ip routing
Use case	Small/medium networks, limited VLANs	Medium/large networks, multiple VLANs

Remember:

Inter-VLAN routing enables communication between different VLANs.

Use Router-on-a-Stick for simple setups and Layer 3 switches for better performance and scalability.

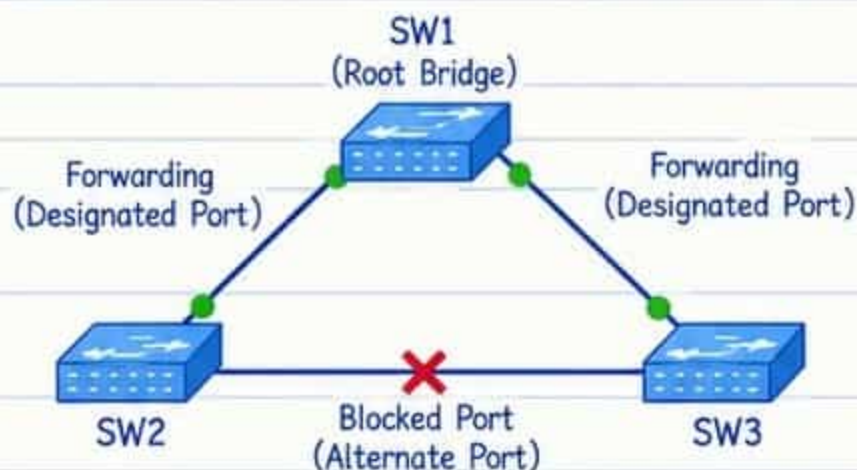
11. Spanning Tree Protocol (STP)

1. Why STP is needed?

- Layer 2 networks can have loops due to redundant links.
- Loops cause broadcast storms (endless frames).
- MAC address table becomes unstable (flapping).
- Duplicate frames can reach the destination.
- STP prevents loops by logically blocking redundant paths.

2. Basic STP working

- STP allows only one active path between switches.
- It blocks redundant links to prevent loops.
- If the active link fails, the blocked link is unblocked and used as a backup.
- STP continuously monitors the network and recalculates the topology if changes occur.



3. Root Bridge election

- The switch with the lowest Bridge ID (BID) becomes the root bridge.
- Bridge ID = Priority (default 32768) + MAC address.
- If priorities are the same, the switch with the lowest MAC address wins.
- You can manually set a switch as root using a lower priority (e.g. 4096).

4. Port roles

- Root Port : Best path to the root bridge (one per non-root switch).
- Designated Port : Best path towards a segment (one per segment).
- Alternate/Blocked Port : Redundant port kept in blocking state to prevent loops.

5. Port states summary

State	Purpose	Forwards Frames?	Duration (default)
Blocking	Prevents loops, does not learn MAC addresses	No	N/A
Listening	Listens for BPDUs, prepares to learn MAC addresses	No	15 seconds
Learning	Learns MAC addresses (builds MAC table)	No	15 seconds
Forwarding	Normal operation, forwards user traffic	Yes	N/A
Disabled	Port is administratively down or not in use	No	N/A

6. STP vs RSTP comparison

Feature	STP (802.1D)	RSTP (802.1w)
Convergence time	Slower (30–50 seconds)	Faster (1–6 seconds)
Port states	5 states	3 states (Discarding, Learning, Forwarding)
Operation	Basic loop prevention	Faster convergence and quicker topology changes
Compatibility	Older standard	Backward compatible with STP
Use case	Common in older networks	Widely used in modern networks

7. CLI / Configuration

```
SW1(config)# spanning-tree vlan 10 root primary    ! Make this switch root for VLAN 10
SW1# show spanning-tree vlan 10                  ! Verify STP information
```

Remember: STP prevents Layer 2 loops by logically blocking redundant paths.

12. EtherChannel

1. What is EtherChannel?

- EtherChannel bundles multiple physical links between two devices into one logical link (port-channel).
- It treats multiple links as a single link, increasing bandwidth and providing redundancy.
- Commonly used between switches, but can also be used between a switch and a router.

2. Benefits of EtherChannel

- **Redundancy:** If one link fails, traffic continues over the remaining links.
- **Load balancing:** Traffic is distributed across multiple links.
- **Higher bandwidth:** Increases total available bandwidth.
- **Simplified STP:** Multiple links appear as a single logical link to Spanning Tree Protocol (STP).

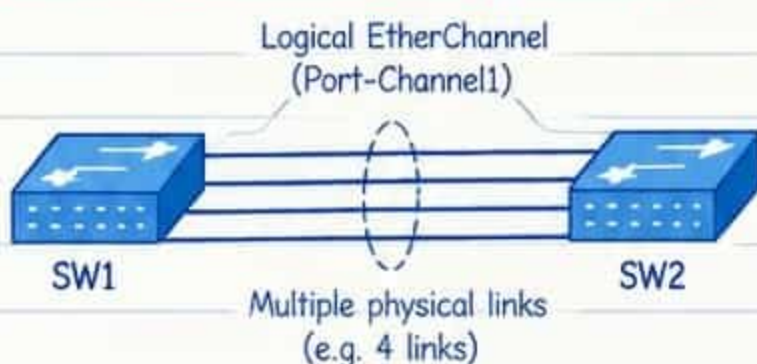
3. EtherChannel Requirements

- Links must have the same speed and duplex settings.
- Links must be the same media type (all copper or all fiber).
- Links should be in the same VLAN (for access) or same trunk settings (for trunk).
- Both ends must use the same EtherChannel protocol (PAgP, LACP, or static "on").
- All links in the bundle must have the same switchport configuration.

4. PAgP vs LACP vs Static "on"

Feature	PAgP	LACP	Static "on"
Protocol	Cisco proprietary	IEEE 802.3ad (standard)	No negotiation
Modes	auto / desirable	passive / active	on (forced)
Negotiation	Yes	Yes	No
Device support	Cisco only	Multi-vendor (standard)	Works with any device
Best use case	Cisco networks	Mixed vendor / standard	Simple, no negotiation needed

5. EtherChannel Topology Example



6. Configuration Example (LACP)

SW1 Configuration (LACP)

```
SW1(config)# interface range g0/1 - 4
SW1(config-if-range)# channel-group 1
mode active
SW1(config-if-range)# exit
SW1(config)# interface port-channel 1
SW1(config-if)# switchport mode trunk
```

SW2 Configuration (LACP)

```
SW2(config)# interface range g0/1 - 4
SW2(config-if-range)# channel-group 1
mode active
SW2(config-if-range)# exit
SW2(config)# interface port-channel 1
SW2(config-if)# switchport mode trunk
```

7. Verification Commands

```
SW1# show etherchannel summary    # Shows EtherChannel status and member links
SW1# show interfaces trunk       # Verifies trunk settings (if using trunk)
SW1# show running-config | include channel-group  # Shows EtherChannel configuration
```

8. Common Issues / Mistakes

- Mismatched link speed/duplex.
- Interfaces in different VLANs or trunk settings.
- Using different EtherChannel protocols.
- One side in active and the other in passive (for LACP).
- Forgetting to configure the port-channel interface.
- Physical links not working or not all added to the bundle.

Remember: EtherChannel bundles multiple physical links into one logical link (port-channel). It provides higher bandwidth, redundancy and simplified STP.

13. Static Routing

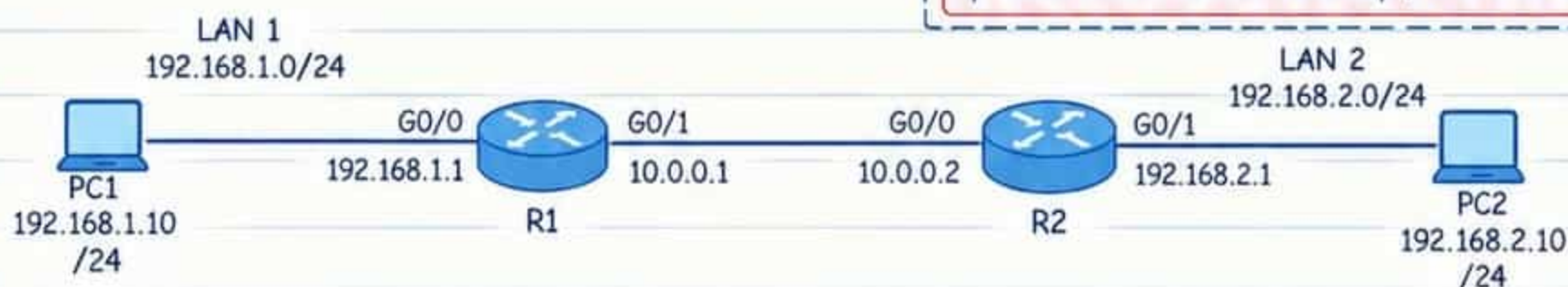
What is Static Routing?

- Static routing is a method where routes are manually configured by the network administrator.
- The router uses a fixed route to reach remote networks.
- Unlike dynamic routing, static routes do not automatically adjust to network changes.

When to Use Static Routes?

- Small networks with few routers.
- When the network topology is stable and does not change often.
- To provide a default route to the internet.
- As a backup route (floating static route).
- In stub networks (networks with a single exit point).

Network Diagram



Static Route Examples

1. Standard Static Route (on R1 to reach LAN 2)

```
R1(config)# ip route 192.168.2.0 255.255.255.0 10.0.0.2
```
2. Default Route (on R2 to send all unknown traffic to R1)

```
R2(config)# ip route 0.0.0.0 0.0.0.0 10.0.0.1
```
3. Floating Static Route (backup route with higher AD)

```
R1(config)# ip route 192.168.2.0 255.255.255.0 10.0.0.2 5
```

(Default administrative distance is 1, so 5 makes it a backup route)

Verification Commands

- Check routing table:

```
R1# show ip route
```
- Test connectivity:

```
R1# ping 192.168.2.10
```
- Trace the path:

```
R1# traceroute 192.168.2.10
```

Advantages vs Disadvantages

Feature	Advantages	Disadvantages
Configuration	Simple and easy to configure	Manual configuration required
Resource Usage	Uses less CPU and memory	Does not adapt to network changes
Predictability	Provides fixed and predictable paths	No automatic route discovery
Scalability	Good for small networks	Does not scale well in large networks
Use Case	Small, stable networks or specific routes	Not suitable for large, dynamic networks

Remember:

Static routes are simple and predictable but do not scale well.
Use them in small, stable networks or for specific purposes like default or backup routes.

14. OSPF Basics

What is OSPF?

- OSPF (Open Shortest Path First) is a link-state dynamic routing protocol.
- It is an interior gateway protocol (IGP) used within an autonomous system (AS).
- It dynamically learns routes and builds a topology map of the network.
- OSPF uses the Shortest Path First (SPF) algorithm to calculate the best path.

Why is Dynamic Routing Useful?

- Automatically learns and updates routes when the network changes.
- Reduces manual configuration compared to static routing.
- Adapts quickly to failures and finds alternate paths.
- More scalable and suitable for medium to large networks.

OSPF Characteristics

- **Link-State:** Shares link-state information with neighbors and builds a topology database.
- **Cost Metric:** Uses cost (based on interface bandwidth) to determine the best path.
- **Fast Convergence:** Quickly responds to network changes.
- **Classless:** Supports VLSM and is classless (carries subnet mask information).

Neighbor / Adjacency

- Routers must be directly connected and in the same area to become neighbors.
- They exchange OSPF messages to build adjacency and synchronize the database.
- Once adjacent, routers share link-state information (LSAs).



R1 and R2 become neighbors on the 10.0.12.0/24 network. R2 and R3 become neighbors on the 10.0.23.0/24 network.

OSPF Packet Types

Type	Purpose
Hello	Discovers neighbors and maintains adjacency (using periodic hello messages).
DBD	Database Description (Summarizes the LSDB contents).
LSR	Link State Request (Requests specific LSAs).
LSU	Link State Update (Carries LSAs).
LSAck	Link State Acknowledgment (Acknowledges received LSAs).

Basic OSPF Configuration (Single Area)

```

R1(config)# router ospf 1
R1(config-router)# network 10.0.12.0 0.0.0.255 area 0
R1(config-router)# network 192.168.1.0 0.0.0.255 area 0
  
```

- Use the network command to enable OSPF on interfaces and assign them to an area (area 0 is the backbone).

Verification Commands

```

R1# show ip ospf neighbor
# Shows OSPF neighbors and their state

R1# show ip route ospf
# Shows routes learned via OSPF
  
```

Static Routing vs OSPF

Feature	Static Routing	OSPF
Route learning	Manually configured	Dynamically learned
Adaptability	Does not adapt to changes	Automatically adapts to network changes
Scalability	Not suitable for large networks	Scalable for medium/large networks
Administration	Simple for small networks	More complex, but easier to manage at scale
Best use case	Small, stable networks	Medium to large, dynamic networks

Remember: OSPF dynamically learns routes and uses cost to choose the best paths.

15. Access Control Lists (ACLs)

What are ACLs?

- ACLs (Access Control Lists) are used to filter network traffic based on specific rules.
- They allow or deny packets based on information such as source IP, destination IP, protocol or port.
- ACLs are used for security, traffic control and network management.

Standard ACL vs Extended ACL

Feature	Standard ACL	Extended ACL
ACL Number Range	1 – 99 (or 1300 – 1999)	100 – 199 (or 2000 – 2699)
Matching Criteria	Source IP address only	Source IP, destination IP, protocol, port
Granularity	Less specific	More specific
Use case	Simple filtering (e.g. block a network)	Advanced filtering (e.g. block a service)
Example	access-list 10 deny 192.168.10.0 0.0.0.255	access-list 101 deny tcp any 192.168.20.0 0.0.0.255 eq 80

Wildcard Mask

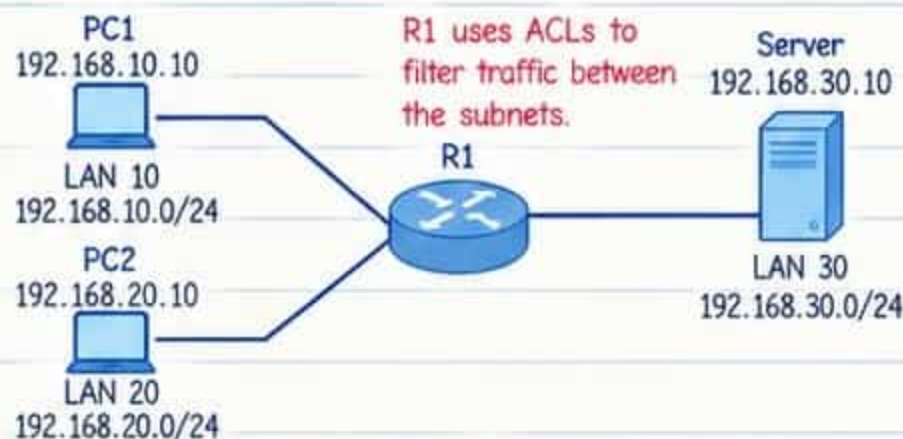
- Wildcard mask is the inverse of a subnet mask.
- 0 = must match, 1 = ignore (any value).
- Used in ACLs to determine which bits to check.

Subnet Mask	Wildcard Mask	Matches
255.255.255.255	0.0.0.0	Single IP address
255.255.255.0	0.0.0.255	All IPs in a /24 network
255.255.0.0	0.0.255.255	All IPs in a /16 network
0.0.0.0	255.255.255.255	Any IP address (any)

Placement Rules

- **Standard ACLs:** place as close to the destination as possible (to avoid blocking other traffic).
- **Extended ACLs:** place as close to the source as possible (to stop unwanted traffic early).

Network Diagram



Example: Standard ACL

```
! Block LAN 10 from accessing LAN 30
R1(config)# access-list 10 deny 192.168.10.0 0.0.0.255
R1(config)# access-list 10 permit any
R1(config)# interface g0/1
R1(config-if)# ip access-group 10 out
```

Example: Extended ACL

```
! Block HTTP from LAN 20 to server 192.168.30.10
R1(config)# access-list 101 deny tcp 192.168.20.0 0.0.0.255
                                     host 192.168.30.10 eq 80
R1(config)# access-list 101 permit ip any any
R1(config)# interface g0/0
R1(config-if)# ip access-group 101 in
```

Verification Commands

- `show access-lists` → Displays ACLs and match counters
- `show ip interface` → Shows which ACLs are applied to interfaces
- `show running-config | include access-list` → View ACL configuration

Common Mistakes

- Forgetting the implicit deny at the end.
- Placing the ACL in the wrong direction or location.
- Using the wrong wildcard mask.
- Not adding a permit statement for required traffic.

Remember:

- ACLs filter traffic based on rules (permit or deny).
- ACLs are processed from top to bottom (first match is used).
- There is an implicit deny any at the end of every ACL.

16. NAT and PAT

Why is NAT Needed?

- Conserves public IPv4 addresses by allowing private IP addresses to access the internet.
- Hides internal private IP addresses from the outside network (adds a layer of security).
- Allows multiple devices in a private network to share a single public IP address (using PAT).
- Helps organizations use private addressing (RFC 1918) internally while accessing external resources.

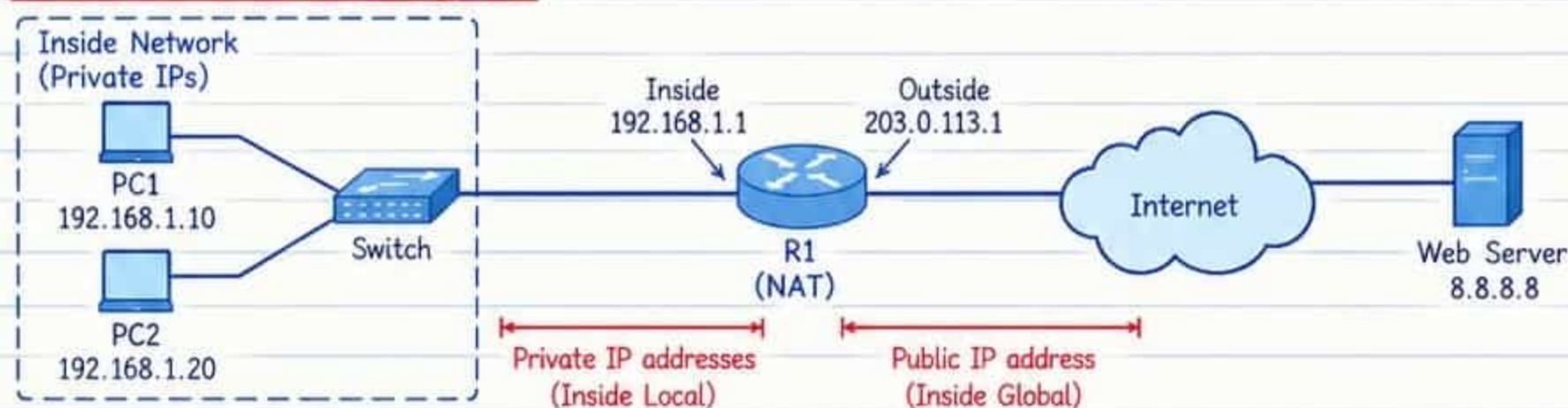
NAT Address Types

- Inside Local - Private IP address of an inside host (as seen from inside).
- Inside Global - Public IP address that represents the inside host (as seen from outside).
- Outside Local - IP address of an outside host (as seen from inside, usually the same as outside global).
- Outside Global - Public IP address of an outside host (as seen from outside).

Static NAT vs Dynamic NAT vs PAT

Feature	Static NAT	Dynamic NAT	PAT (NAT Overload)
Mapping type	One-to-one (fixed)	Many-to-many (from a pool)	Many-to-one (using ports)
Public IP usage	One public IP per host	Pool of public IP addresses	Single public IP address
Configuration	Static mapping	Dynamic pool + ACL	Overload (uses port numbers)
Best use case	Servers (e.g. web, mail)	Multiple hosts, limited pool	Most common, Internet access
Conserves IPs	No	Yes (to some extent)	Yes (highly efficient)

NAT/PAT Network Diagram



NAT/PAT Configuration Example

```

R1(config)# access-list 1 permit 192.168.1.0 0.0.0.255
R1(config)# interface g0/0
R1(config-if)# ip nat inside
R1(config-if)# exit
R1(config)# interface g0/1
R1(config-if)# ip nat outside
R1(config-if)# exit
R1(config)# ip nat inside source list 1 interface g0/1 overload

```

Verification Commands

```

R1# show ip nat translations
! Displays active NAT translations

R1# show ip nat statistics
! Shows NAT statistics and counters

R1# show running-config | include ip nat
! Shows NAT-related configuration

```

Benefits and Limitations

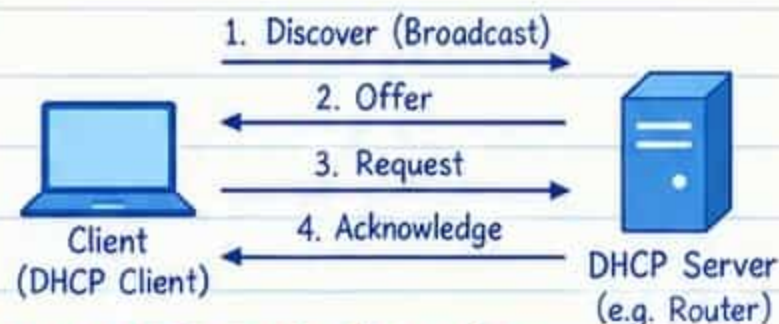
Benefits	Limitations
• Conserves public IPv4 addresses • Provides basic security (hides internal IPs) • Allows private networks to access the internet • Easy to implement (especially PAT)	• Does not provide strong security (use a firewall) • Some applications may have issues (e.g. IPsec, VoIP) • Breaks end-to-end connectivity • Requires additional configuration and management

Remember: PAT (NAT Overload) lets many private hosts share one public IP address using port numbers to differentiate between connections.

17. DHCP and WLAN Basics

1. DHCP (Dynamic Host Configuration Protocol)

- DHCP automatically assigns IP addresses and other network settings to devices (clients).
- It reduces manual configuration and helps in centralized management of IP addressing.
- DHCP uses a client-server model and typically runs on UDP ports 67 (server) and 68 (client).



The DORA Process

- **Discover** – Client broadcasts a request for an IP address (0.0.0.0 to 255.255.255.255).
- **Offer** – Server responds with an available IP address and other information.
- **Request** – Client requests the offered IP address.
- **Acknowledge** – Server confirms the lease and provides the IP configuration.

Important DHCP Fields / Benefits

- Assigned IP address (leased for a specific time)
- Subnet mask
- Default gateway
- DNS server
- Lease time (how long the IP is valid)
- Simplifies network management
- Reduces IP address conflicts
- Efficient use of IP address space

DHCP Pool Configuration Example (on a Router)

```

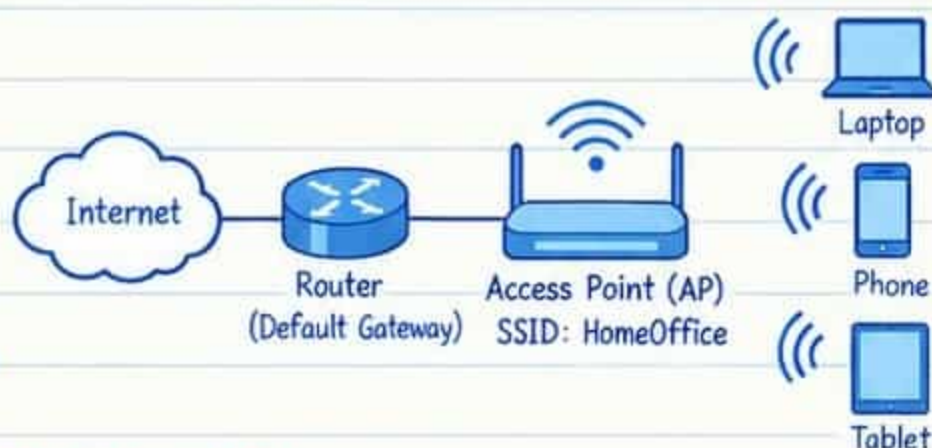
R1(config)# ip dhcp pool LAN
R1(dhcp-config)# network 192.168.10.0 255.255.255.0
R1(dhcp-config)# default-router 192.168.10.1
R1(dhcp-config)# dns-server 8.8.8.8
R1(dhcp-config)# lease 7
  
```

Static IP vs Dynamic IP (DHCP)

Feature	Static IP	Dynamic IP (DHCP)
Assignment	Manually configured	Automatically assigned
Changes	Does not change	May change (after lease)
Management	Higher effort	Lower effort
Best for	Servers, network devices	End users, client devices
Flexibility	Less flexible	More flexible

2. WLAN (Wireless Local Area Network)

- A WLAN provides wireless network access to devices using radio waves instead of cables.
- It allows mobile devices (laptops, smartphones, etc.) to connect to a network.
- A Wireless Access Point (AP) connects wireless clients to the wired network.
- Devices connect to a WLAN using a network name called the SSID (Service Set Identifier).



WLAN Key Concepts

- **SSID** – Unique name of the wireless network (e.g. HomeWiFi).
- **Access Point (AP)** – Provides wireless connectivity and bridges wireless clients to the wired network.
- **Frequency Bands** – WLAN typically operates on 2.4 GHz and 5 GHz bands.
- **Security** – Use WPA2 or WPA3 to encrypt wireless traffic and protect the network from unauthorized access.

2.4 GHz vs 5 GHz

Feature	2.4 GHz	5 GHz
Range	Longer range	Shorter range
Speed	Lower speeds	Higher speeds
Interference	More interference (microwaves, Bluetooth, etc.)	Less interference
Number of channels	Fewer (3 non-overlapping)	More channels
Best for	Larger area, basic needs	Higher performance, less crowded areas

WPA2 vs WPA3

- **WPA2** – Widely used, uses AES encryption. Still secure but older standard.
- **WPA3** – Newer and stronger security, better protection against attacks, especially for personal networks.

Remember: DHCP automates IP assignment and WLAN provides wireless access securely.

18. Network Troubleshooting Basics

1. Troubleshooting Approach



2. Common Tools / Commands

Command / Tool	Purpose	Example
ping	Tests reachability to a device	ping 192.168.1.1
tracert (tracert)	Shows the path to a destination	tracert 8.8.8.8
ipconfig / ifconfig	Displays IP configuration on a host	ipconfig (Windows) / ifconfig (Linux)
show ip interface brief	Checks interface status and IP addresses	show ip interface brief
show running-config	Displays the current configuration	show running-config
show vlan brief	Shows VLANs on a switch	show vlan brief
show ip route	Displays the routing table	show ip route

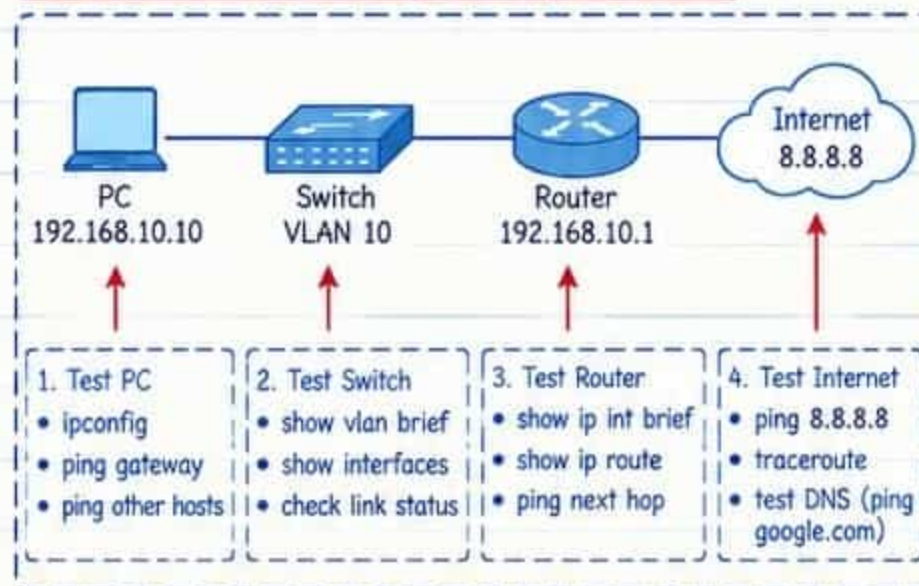
3. Common Network Issues

- Wrong IP address – device has incorrect IP, subnet mask or gateway.
- Cable problem – faulty or unplugged cable.
- Duplex mismatch – different duplex settings (e.g. one side full, other half).
- VLAN mismatch – device in wrong VLAN.
- Down interface – interface is administratively down or physically down.
- Routing issue – missing or incorrect route.
- DNS issue – cannot resolve domain names.

4. OSI-Based Troubleshooting

7	Application	– Check application (e.g. web, DNS, email)
6	Presentation	– Usually not an issue in CCNA
5	Session	– Usually not an issue in CCNA
4	Transport	– Check TCP/UDP ports (e.g. ping, traceroute)
3	Network	– Check IP addressing, routing, ACLs
2	Data Link	– Check VLAN, switching, MAC address, duplex
1	Physical	– Check cables, link lights, physical connections

5. Where to Test? (Example Path)



Remember:

Start simple – check the basics first.
 Verify Layer 1 (physical) before moving to higher layers.
 Use a logical, step-by-step approach to find and fix the problem.